RETIREPILOT

Technical Specification v2.7

Actual code architecture, data and API contracts, test evidence, controls and unresolved gaps

DECISION STATUS / APPROVED PRODUCT CHOICES 1A-7A | PILOT RELEASE / NO-GO

Effective date: 4 November 2026. Supersedes PSD/TSD v2.0 on all changed scope, controls and implementation-status statements. This specification does not constitute financial, security or legal certification.

1. Technical scope and version provenance

Built from the RetirePilot v2.6 local package, with v2.7 changes backed by tests. Python/FastAPI deterministic calculations remain the default. No production deployment, secure customer accounts, CPF connection, verified consent provider, trained model, external action execution or authenticated verification service exists.

Component	Current module	Status
Household schema	retirepilot/core.py	Synthetic validated input; not production consent
Monthly cash flows	retirepilot/projection_v22.py	P0 partial-month denominator repaired; bounded assumptions
Classifier	classifier_legacy.py / adapter.py	96 evaluated internally; provisional display only
Consumer assessment	service.py / api.py / ui.html	Redacted persona view; localhost research API
CPF reference	cpf_retirement_v26.py; statutory_2026.py; policy.py	Bounded; no full statutory lifetime calculator
Scenario design	catalog_scenarios_v26.py / priority_scenarios_v27.py	46 contracts; 5 priorities, 4 bounded previews, 1 blocked
Synthetic local vault	local_vault_v24.py	Encrypted at rest; sharing always blocked; no verified identity
Assistant	assistant.py / planner_session_v24.py	Rule-based; no live LLM; no regulated recommendations

2. Data contracts and trust boundaries

- Household: id, as_of, primary_id, persons, explicit income_streams/assets/liabilities/expenses, goal, assumptions. Missing data is unknown, not zero.
- Person: id, role, dob, residency, optional CPF OA/SA/MA/RA balances. Adult financial fields must not be shared without authenticated person-specific consent.
- CPF LIFE income stream: type=cpf_life, start age, amount, owner, and source from member_transcribed_official_cpf_estimate, or synthetic_user_provided_estimate only when profile is synthetic.
- Source markers are member-declared and not cryptographic CPF proof. Real-person identity verification is not connected. No public API should imply authentication.
- Income streams include reported cash deductions, worker CPF and tax reserves; they are not inferred statutory deductions.

3. Exact partial-month correction (P0)

Former bug: when the last month was truncated at a projection-end birthday, the end date also became the denominator; the engine treated the remaining days as one full month. The corrected fraction divides included calendar days by the number of days in the entire calendar month.

time_fraction = (exclusive_period_end - inclusive_period_start).days / days_in_calendar_month(year, month). The same denominator applies to clipped income or expense period intersections. Month-level start and end are exclusive at the birthday boundary.

Independent reference case	Expected result	Automated verification
S$7,200 / month, Jan 1 to May 20 2027 (exclusive)	S$33,212.90 = S$7,200 x (4 + 19/31)	PASS, tests/test_approved_decisions_v27.py
Birth anniversary May 1, horizon ends May 1	S$28,800 for Jan-Apr	PASS
S$3,100 / month expense, same 19-day May partial	S$14,300 = S$3,100 x (4 + 19/31)	PASS

4. Persona display separation (3A)

The internal deterministic classifier still computes observed scores, coverage and weights for traceability. Consumer-facing service.public_persona_summary returns only evaluated count, rule version, descriptive top-three ids/names/status and a non-confidence notice; it deliberately excludes all_results, raw scores, numeric thresholds and evidence coverage. The UI no longer creates score bars, percentages or confidence labels. Direct developer classifier imports are not a consumer interface and must remain access-controlled if ever deployed.

5. CPF LIFE estimate provenance (2A)

core.validate rejects any reported personal CPF LIFE income without a valid source declaration. A member_transcribed_official_cpf_estimate is a self-declaration of the official estimate; no API-based verification exists. In synthetic fixtures, synthetic_user_provided_estimate is permitted. Public CPF Board tables may be used for separately labelled examples, never as personalised income. Payment timing must satisfy existing age constraints.

Open: no live Singpass/CPF connector, no authenticity checking, no comprehensive CPF interest/RA lifecycle, no personalised actuarial model, no tax/CPF legal determination. The product must not present these as available.

6. Household identity and consent fail-closed state (4A)

LocalVault still supports purpose-specific synthetic consent for local development. The new share_household_records interface always raises a denial, regardless of caller-provided strings or mock evidence. A secure production architecture requires an independent provider to authenticate each adult and prove informed consent, ownership checks, revocation, purpose limitation, audit and third-party security review. The synthetic checkbox cannot meet this gate.

7. Five priority event interfaces (5A)

Scenario	Required inputs	Return and unsupported scope
S020 downsizing	sale price, costs, outstanding mortgage, official member-supplied CPF property refund, replacement price	Cash proceeds sensitivity; no housing eligibility, tax/CPF accounting or simultaneous mortgage switch
S030 payout timing	member-provided official monthly estimates for ages 65 and 70	Monthly difference only, no break-even or personal projection
S004 part-time	current and proposed net monthly earned cash, duration months	Net cash sensitivity; future tax/CPF re-estimation omitted
S016 parent care	incremental monthly caregiving cash expense, number of months	Cost sensitivity; no subsidy, caregiver wage-loss or benefit eligibility
S012 divorce	none accepted in prototype	Blocked until qualified legal inputs and verified household consent

Priority_scenarios_v27 previews explicitly return may_post_to_household_projection=false. These calculations do not update the household plan or count as a complete scenario operator. The original 46-entry catalogue remains unchanged.

8. API surface and release control endpoint

Route	Scope	Safety limitations
GET /	Local UI	Browser not penetration-tested
GET /api/sample	Synthetic sample only	No secrets or real-person data
POST /api/evaluate	Conditional snapshot, redacted persona and projection	Synthetic prefix can be spoofed; never expose publicly
POST /api/ask	Rule-based educational assistant	Model disabled by default; not regulated advice
POST /api/reference	Bounded statutory/reference calculators	Not personal eligibility results
GET /api/release-controls	Read-only controls status	Sharing blocked, identity unintegrated, Phase 9 NO-GO

9. Tests and reproducibility

Local workspace (synthetic data only): 609/609 pytest cases, 498/498 legacy persona contracts, and 13/13 existing API checks. These cover the new partial-month and approved-decision boundary tests. The 13 legacy API checks do not independently verify the new release-controls route; that route is tested using FastAPI TestClient in pytest.

Run from project root: python -m pip install -r requirements.txt; python -m pytest -q; PYTHONPATH=. python audit/run_rule_contracts_v22.py; python audit/run_api_checks.py. For UI testing: uvicorn retirepilot.api:app --host 127.0.0.1 --port 8766.

No GitHub CI run or independent third-party acceptance is claimed. Legacy tests are mostly generated rule contracts, not proof of classifier precision. Browser usability and WCAG verification still outstanding.

10. Threats and unresolved technical work

- P0: prohibit real-person data, Internet exposure, unauthenticated multi-adult household sharing and use as personal financial advice until security/legal gates pass.

- P0: independently reconcile full CPF contribution pathways, account interest and transfers, CPF-funded housing, tax and future CPF LIFE representations.
- P0: integrate authenticated identity, household roles, verified adult consent, key management, audit, deletion, rate limits, TLS and threat mitigation.
- P1: add actual source provenance capture and member consent UX; current source flag is self-declared, not official verification.
- P1: complete five priority scenarios with real temporal modelling, then remaining catalogue entries, with independent golden cases.
- P1: calibrate 96 personas with blinded, representative labels and fairness testing; hide scores until approved.
- Release: external financial, security and qualified Singapore legal sign-offs; 48 real consumer sessions. No shortcuts based on hypothetical PASS.

11. Integration and change-control status

The approved decisions are committed to private GitHub documentation. Local v2.7 source changes are packaged separately; the full codebase remains awaiting remote migration and CI. GitHub documentation commits must not be mistaken for merged code, passed remote CI or an authorised pilot. Product-spec decisions govern future implementation, not evidence of completed production capability.

Authoritative reference links

https://www.cpf.gov.sg/service/article/when-can-i-start-receiving-my-cpf-life-payouts

https://www.cpf.gov.sg/service/article/how-much-cpf-payouts-can-i-get-every-month

https://www.cpf.gov.sg/service/article/what-is-the-monthly-payout-estimator

https://www.mas.gov.sg/regulation/guidelines/guidelines-on-provision-of-financial-advisory-service

https://www.pdpc.gov.sg/